**Project Report: Sentiment Classification Using Supervised Learning**

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**1. Problem Statement**

The problem chosen for this project is **Sentiment Classification of Text Data** using supervised machine learning. The goal of sentiment classification is to automatically determine whether a given piece of text expresses a **positive, negative, or neutral** sentiment.

With the growth of online reviews, social media, and customer feedback, organizations need automated systems to analyze opinions and feedback efficiently. Manual analysis of large amounts of text data is time-consuming and inefficient, which makes sentiment analysis an important application of machine learning and natural language processing.

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**2. Why This Problem Matters**

Sentiment analysis is widely used in real-world applications such as:

- Product review analysis
- Customer feedback analysis
- Social media monitoring
- Market research
- Brand reputation management
- Movie and product recommendation systems

Companies use sentiment analysis to understand customer satisfaction, improve products, and make better business decisions. Therefore, building a sentiment classification model is a practical and important machine learning problem.

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**3. Approach to Solving the Problem**

To solve the sentiment classification problem, a **Supervised Machine Learning approach** was used. In supervised learning, the model is trained on labeled data where each text already has a sentiment label (positive, negative, neutral).

**Steps Followed in the Project**

1. Create dataset of text and sentiment labels
2. Convert text data into numerical features using TF-IDF Vectorization
3. Split data into training and testing sets
4. Train a machine learning model (Logistic Regression)
5. Test the model on unseen data
6. Evaluate the model using Classification Report and Confusion Matrix

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**4. Dataset Description**

A small dataset was created manually containing sample text reviews and their corresponding sentiment labels.

The dataset contains two columns:

- **Text** – Review or sentence
- **Sentiment** – Positive, Negative, or Neutral

Example:

| Text | Sentiment |
|---|---|
| I love this product | Positive |
| Worst purchase ever | Negative |
| It is okay | Neutral |

This dataset was converted into a Pandas DataFrame for processing.

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**5. Data Preprocessing**

Machine learning models cannot understand text directly, so text data must be converted into numerical form.

**TF-IDF Vectorization**

TF-IDF (Term Frequency – Inverse Document Frequency) is used to convert text into numerical vectors based on word importance.

In this project:

- Stop words were removed
- Unigrams and Bigrams were used
- Text data was converted into TF-IDF feature vectors

This step transforms text into a matrix that the machine learning model can understand.

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**6. Model Used**

The machine learning model used in this project is **Logistic Regression**.

**Why Logistic Regression?**

- Works well for text classification
- Simple and efficient

- Good baseline model for NLP problems
- Works well with TF-IDF features
- Easy to train and interpret

The model was trained using the training dataset and then used to predict sentiment on the test dataset.

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**7. Model Evaluation**

After training the model, predictions were made on the test data and evaluated using:

**Classification Report**

It shows:

- Precision
- Recall
- F1-Score
- Accuracy

**Confusion Matrix**

It shows how many predictions were:

- Correctly predicted
- Incorrectly predicted
- Misclassified between positive, negative, and neutral

These metrics help evaluate the performance of the model.

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**8. Key Decisions Made**

During the project, the following key decisions were made:

- Use **TF-IDF Vectorizer** instead of Bag of Words
- Use **Logistic Regression** instead of complex models
- Use **Train-Test Split (80-20)** for evaluation
- Use **N-grams (1,2)** to capture word combinations
- Remove **stop words** to improve model performance

These decisions helped improve the quality of the sentiment classification model.

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**9. Challenges Faced**

Some challenges faced during the project include:

- Small dataset size
- Text preprocessing and vectorization
- Choosing the right machine learning model
- Understanding evaluation metrics
- Handling neutral sentiment classification

Because the dataset was small, the model accuracy may not be very high, but the main goal was to understand the process of sentiment classification.

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**10. What I Learned**

From this project, I learned:

- Basics of **Natural Language Processing (NLP)**
- How **TF-IDF Vectorizer** works
- Difference between **training data and testing data**
- How **Logistic Regression** is used for classification
- How to evaluate machine learning models
- Importance of **data preprocessing**
- End-to-end machine learning workflow

This project helped me understand how machine learning can be used to solve real-world text classification problems.

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**11. Conclusion**

In this project, a sentiment classification system was built using supervised machine learning techniques. Text data was converted into numerical features using TF-IDF vectorization, and a Logistic Regression model was trained to classify sentiments into positive, negative, and neutral categories.

The project demonstrates the basic workflow of a machine learning project, including data preparation, feature extraction, model training, prediction, and evaluation. This project can be further improved by using a larger dataset, deep learning models, and advanced NLP techniques.

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**12. Future Improvements**

The project can be improved by:

- Using a larger dataset
- Using Deep Learning models like LSTM or BERT

- Performing advanced text preprocessing
- Hyperparameter tuning
- Deploying the model as a web application
- Real-time sentiment analysis system

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**End of Project Report**